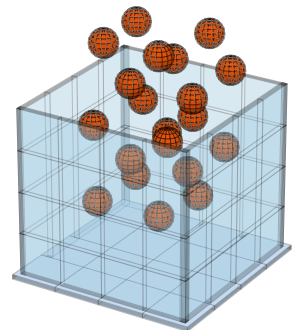
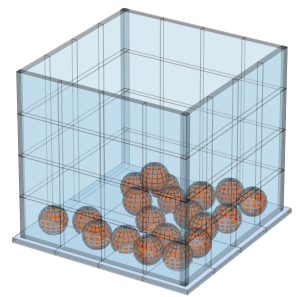


Test scenes. (a) Soft clutter: 20 spheres ($r = 2.5$ cm, water density) dropped into a 30 cm bin, $k = 10^3$ N/m, $v_s = 1$ cm/s. (b) Hard clutter: 10 spheres + 10 cubes ($h = 2.5$ cm), $k = 10^5$ N/m, $v_s = 0.1$ mm/s. (c) Bouncing ball: 0.1 kg, $r = 5$ cm, $k = 10^3$ N/m, zero dissipation, dropped from 1 m.

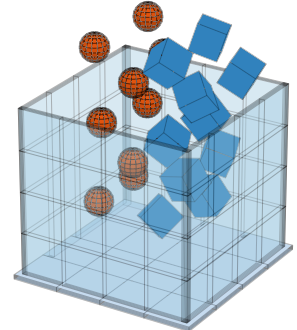
(a) Soft clutter
 $t = 0$



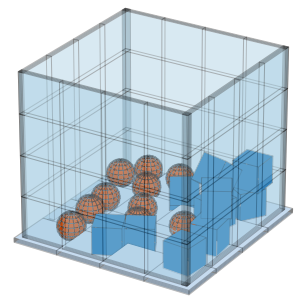
$t = 1.5$ s (ICF error control, $\epsilon = 10^{-3}$)



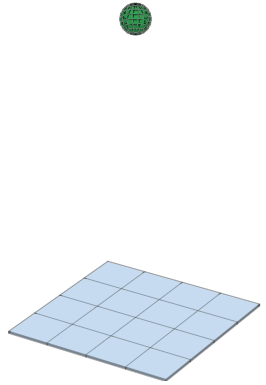
(b) Hard clutter
 $t = 0$



$t = 1.5$ s (ICF error control, $\epsilon = 10^{-3}$)



(c) Bouncing ball
 $t = 0$



$t = 0.45$ s (ICF error control, $\epsilon = 10^{-3}$)

